

Problem 25 (Italy MO 2016 Problem 2). In a mathematical contest, there are 3 problems. A contestant can get an integer score between 0 and 7 (including 0 and 7) in each problem. It is known that for any two contestants, there exists at most one problem in which they have obtained the same score. For example, there are no two contestants whose scores are 6, 7, 0 and 6, 7, 1 in this order, but there might be two contestants whose scores are 6, 7, 0 and 7, 6, 0 in this order.

Find the maximum number of contestants.

Solution. The answer is 64.

Suppose there are at least 65 contestants. The scores obtained by a contestant in the first two problems only have $8 \times 8 = 64$ possibilities. By the pigeonhole principle, two contestants must have the same scores in the first two problems. This is a contradiction. So there can be at most 64 contestants.

Here gives an example for which there are 64 contestants. Suppose the scores of the contestants are (a_i, b_i, c_i) , where (a_i, b_i) runs through all 64 different possible pairs of integers between 0 and 7, and c_i is the unique integer between 0 and 7 satisfying $8 \mid a_i + b_i + c_i$. If two of a_i, b_i, c_i are the same as the corresponding two of a_j, b_j, c_j , then the third is also the same by $a_i + b_i + c_i \equiv a_j + b_j + c_j \pmod{8}$. This implies $i = j$. So the condition is satisfied. $\square$

Problem 26 (Saudi Arabia GMO TST 2015 Problem 2). There is a finite number of stones. The weight of each stone is not greater than 0.5kg, and the total weight of the stones is not greater than 2.5kg. Prove that the stones can be divided into 4 groups, so that each group has a total weight not greater than 1kg.

Solution. We simply need to put the stones in the 4 groups one by one in an arbitrary way, as long as the total weight of stones in each group does not exceed 1kg. We claim that we can always do so, until all stones have been put. Suppose on the contrary that a stone with weight w kg cannot be put. Then each group has total weight larger than $(1 - w)$ kg. So we have $2.5 > w + 4(1 - w) = 4 - 3w$. This yields the contradiction $w > 0.5$. So we are done. $\square$

Problem 27 (International Zhautykov Olympiad 2015 Problem 5). Let A_n be the set of all partitions of the sequence $1, 2, \dots, n$ into subsequences such that every two adjacent terms of each subsequence have different parities. Let B_n be the set of all partitions of the sequence $1, 2, \dots, n$ into subsequences such that all the terms of each subsequence have the same parity. For example, the partition $(1, 4, 5, 8), (2, 3), (6, 9), (7)$ is an element of A_9 , and the partition $(1, 3, 5), (2, 4), (6)$ is an element of B_6 .

Prove that for every positive integer n , the sets A_n and B_{n+1} contain the same number of elements.

Solution. We say that two integers in a partition (of A_n or B_{n+1}) are *linked* if they are adjacent terms in the same subsequence. For example, in the partition $(1, 2, 3, 4), (5)$ of A_5 , the pairs $(1, 2)$, $(2, 3)$ and $(3, 4)$ are linked. We define a mapping from A_n to B_{n+1} as follows. For any partition p_A in A_n , we define its image p_B (a partition of B_{n+1}) by forming the linkage such that two integers $c < d$ are linked in p_A iff the integers c and $d + 1$ are linked in p_B . For example, the image of the partition $(1, 2, 3, 4), (5)$ is $(1, 3, 5), (2, 4), (6)$, because the pairs $(1, 3)$, $(2, 4)$ and $(3, 5)$ are linked. We now show that this is a bijection.

Firstly, the links in partitions of A_n join two integers of distinct parities. Since $c \not\equiv d \pmod{2}$ implies $c \equiv d + 1 \pmod{2}$, the links in the image join two integers of the same parity. Secondly, each integer in the new linkage can only be linked to at most one integer larger than it (if both $c < d_1 + 1$ and $c < d_2 + 1$ are linked in the image, then both $c < d_1$ and $c < d_2$ are linked originally, contradiction), and linked to at most one integer smaller than it. So the image corresponds to some partition in B_{n+1} .

The mapping is a bijection because we can define its inverse easily. We simply need to map the linkage between $(c, d + 1)$ in p_B to the linkage between (c, d) in p_A . The proof goes the same as above.

As there is a bijection between A_n and B_{n+1} , they have the same number of elements. $\square$

Problem 28 (Belarusian MO 2010 Problem 7). Let p be an odd prime. There are n points on the plane with no three collinear. Any pair of points is joined by a segment. Each segment is coloured in one of the $p + 1$ given colours so that each colour is used at least once and the following condition holds: if two sides of a triangle have the same colour, then the third side also has the same colour. Find the maximum value of n in terms of p for which such a colouring is possible.

Solution. The answer is p^2 .

Suppose $n \geq p^2 + 1$. Fix any point A . Among the p^2 segments with A as an endpoint, at least $\left\lceil \frac{p^2}{p+1} \right\rceil = p$ of them are of the same colour by the pigeonhole principle. Suppose $AB_1, AB_2, \dots, AB_k$ are red for some $k \geq p$, and all other segments with A as an endpoint are not red. Note that $B_i B_j$ is red for any $i \neq j$. As all colours are used, there must be some point C such that AC is not red. If one of $CB_1, CB_2, \dots, CB_p$ is red, say CB_1 , then by considering the triple A, C, B_1 , we deduce the contradiction that AC is red. So all segments $CA, CB_1, CB_2, \dots, CB_p$ are of the other p colours. By the pigeonhole principle, two of them have the same colour different from red, which forces the segment between two of $A, B_1, B_2, \dots, B_p$ to be non-red. This is again a contradiction. Therefore, $n \leq p^2$.

Here gives an example for which $n = p^2$. Label the points (a, b) where each of a, b is one of $0, 1, \dots, p-1$. For any two points (a, b) and (c, d) , we colour the segment joining them in colour p if $a = c$. If $a \neq c$, we use colour r where r is the unique integer between 0 and $p-1$ such that $(a-c)r \equiv b-d \pmod{p}$. This is uniquely determined because p is a prime and $(a-c, p) = 1$. Now, if the segments joining $(a, b), (c, d)$ and $(c, d), (e, f)$ are of colour p , we have $a = c = e$ and so the segment joining $(a, b), (e, f)$ is of colour p . If the segments joining $(a, b), (c, d)$ and $(c, d), (e, f)$ are of colour r for some $0 \leq r \leq p-1$, then $(a-c)r \equiv b-d \pmod{p}$ and $(c-e)r \equiv d-f \pmod{p}$. Adding these, we obtain $(a-e)r \equiv b-f \pmod{p}$, and so the segment joining $(a, b), (e, f)$ is of colour r , as desired. $\square$